Apply filters to SQL queries

Project description

Potential security issues that involve login attempts and employee machines have been brought up to be investigated. Using SQL filters to retrieve records from the `employees` and `log_in_attempts` tables will help recover from these incidents and improve the security posture of the organization.

Retrieve after hours failed login attempts

The log_in_attemps table records all log in attempts and when they were made. The success column indicates whether the login was successful or not using a 1 representing a successful attempt and 0 as a failed attempt.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = 0;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  |       0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  |       0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  |       0 |
|      107 | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 |       0 |
|      111 | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   |       0 |
|      127 | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  |       0 |
|      131 | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 |       0 |
|      155 | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 |       0 |
|      160 | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  |       0 |
|      199 | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
19 rows in set (0.000 sec)
```

As seen in the screenshot above, I selected all the contents from the `log_in_attempts` table and used the `AND` operator to find where the log in attempts failed after 6pm which is after office hours. I found that 19 failed attempts were made after this time.

Retrieve login attempts on specific dates

Next I checked the login attempts on May 8th and 9th of 2022 since a suspicious event occurred on the 9th.The screenshot below shows the query I used to investigate this.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

I used the `WHERE` operator in combination with the `OR` operator to check the login attempts on both days. The `OR` operator lets me analyze the attempts on the 8th or the 9th. I found that there were 75 login attempts during the span of both days

Retrieve login attempts outside of Mexico

Investigations by my team were then made to check all login attempts that were made outside of Mexico. I investigated the `country` column which is shown in the following screenshot.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
```

This query uses the `NOT` and `LIKE` operators to search the `country` column. The `NOT` operator will filter all results where an argument is not in place. The `LIKE` operator is used to search for a pattern in a column. In this case the operators were used to find results where the country does not start with the string 'MEX%'. The `%` symbol is a substitute for any number of other characters that can proceed. The result of this search was that 144 login attempts were made from outside of Mexico.

Retrieve employees in Marketing

The next task was to research employees using the `employees` table. Updates to machines have to be made for employees that are in the Marketing department and are located in the East building.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+-------------+----------+------------+----------+
| employee_id | device_id   | username | department | office   |
+-------------+-------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+-------------+----------+------------+----------+
7 rows in set (0.001 sec)
```

Using a combination of `AND` and `LIKE` operations were used to filter out only the employees that match the description. The `%` symbol was used again to find all offices that started with 'EAST%'. Doing this gave me all the information of those employees that needed to get updates to their machines. The first employee has the username `elarson` as seen in the screenshot.

Retrieve employees in Finance or Sales

Another update was due for all the employees in the Finance and Sales departments so a similar query was made to find the correct employees.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
```

The `OR` operator was used to search the `department` column of the `employees` table to find who needs the required updates to their machines. The username of the first employee found in the Sales department was `lrodriqu`.

Retrieve all employees not in IT

The final research that was needed was for a final update that was done for the IT department already. Since this is the case, I needed to find all employees from all other departments either then the Information Technology department.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
```

This query uses the `NOT` operator to filter out all employees that are in the IT department though the `department` column. The result of this query found that there are 161 employees outside of the IT department that need this update to their machine.

Summary

Combining operators to write SQL queries can help filter through datasets and find important information in order to investigate security issues within an organization. Using simple operators including WHERE, AND, OR, NOT and LIKE, can achieve a lot as shown in this examination and will lead to an improved security posture.